

A Comparative Study of YOLOv8–YOLOv12 for Real-Time Emotion Detection

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Abstract—Emotion detection is a rapidly growing area in artificial intelligence that enables machines to interpret human emotional states based on facial expressions. In this paper, we present the application of YOLO (You Only Look Once) models—YOLOv8 through YOLOv12—for real-time emotion detection. Each version brings improvements in architecture, inference time, accuracy, and robustness. The custom-labelled emotion dataset includes images classified into emotional states such as happy, sad, and angry. The system uses these labelled images for training, and the models are then evaluated based on performance metrics including mean Average Precision (mAP), classification accuracy, frames per second (FPS), and memory efficiency. The best-performing model is recommended based on specific application requirements. A single neural network predicts bounding boxes and class probabilities directly from full images in one evaluation. Since the whole detection pipeline is a single network, it can be optimized end-to-end directly on detection performance. **Index Terms** — YOLO, Deep learning, Emotion Detection- tion, Real-Time Processing, Object Discovery, Facial Expression Recognition, Model Comparison

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I. INTRODUCTION

Facial Emotion Recognition (FER) is a quickly growing area in computer vision and artificial intelligence (AI), designed to identify and categorize human emotions from facial expressions using computer algorithms. Over the years, FER techniques have changed from older methods that relied on manually created features [10], like geometric descriptions and statistical models [1], to more modern approaches that use deep learning, such as convolutional neural networks (CNNs) [2], recurrent neural networks (RNNs), and attention mechanisms [11]. These improvements have greatly enhanced the reliability and ability of FER systems to perform well in real-world situations.

FER is important in many different areas. In human-computer interaction, systems that understand emotions can give

more personalized responses, improving user experience in games, virtual reality, and smart assistants [12]. In healthcare, FER systems are used to keep track of patients' emotions, detect stress or depression, and create individualized treatment plans [13]. In education, FER helps tutoring systems adjust teaching methods to better engage students and improve learning results [14]. FER is also important in security and surveillance, where it helps identify suspicious behavior through emotional cues [15]. In marketing and customer analysis, it helps understand consumer preferences through sentiment recognition [16].

Among the methods used for FER, the You Only Look Once (YOLO) framework has become one of the most effective due to its ability to provide fast, accurate, and efficient performance [4]. Unlike traditional two-stage detectors like the R-CNN family, YOLO uses a single, end-to-end architecture [17] to predict both bounding boxes and class probabilities at the same time, making it ideal for real-time FER applications where delays can be a problem.

Since its introduction, the YOLO family has gone through rapid development, with each new version improving upon the previous ones. YOLOv8, developed by Ultralytics [5], places a strong focus on modularity, stable training, and flexibility during deployment. YOLOv9 [6] improved the architecture by adding programmable gradient information, which helps with optimization across different tasks. YOLOv10 [7] enhances scalability and offers real-time end-to-end detection, while YOLOv11 [8] brings better architectural improvements and better balance between model size, accuracy, and speed. The latest version, YOLOv12 [9], integrates attention-based mechanisms to improve detection in complex and dynamic environments, which is essential for FER because subtle emotional cues can be hard to detect under challenging conditions.

Selecting the most appropriate YOLO version for FER depends on multiple parameters, including:

- Accuracy, to distinguish subtle differences between sim-

ilar emotions such as fear and surprise;

- Inference time and frames per second (FPS), essential for real-time FER in interactive systems;
- Model size and computational efficiency, critical for deployment on embedded systems and mobile devices;
- Adaptability to new datasets, as FER involves diverse facial variations across cultures, ages, and contexts [18];
- Generalization capability, ensuring robustness under different lighting conditions, occlusions, and head poses [19].

With the fast progress in YOLO models, a detailed analysis of how they have developed and how well they fit into FER is needed. This paper addresses that by thoroughly examining YOLO versions from YOLOv8 to YOLOv12, focusing on their architectural changes, strengths, and challenges, and how suitable each is for FER tasks.

The remainder of the article is structured as follows: Section 2 provides a comprehensive literature review of YOLO-based models, outlining their progression and key features. Section 3 details dataset preparation, experimental setup, and the training pipeline used in FER. Section 4 presents results and analysis, comparing YOLOv8 through YOLOv12 across accuracy, efficiency, and adaptability metrics. Finally, Section 5 concludes the article, summarizing key findings and suggesting potential future research directions to advance FER further.

II. LITERATURE REVIEW/RELATED WORKS

A. EARLY APPROACHES TO EMOTION DETECTION

Facial emotion recognition (FER) began with traditional machine learning methods. In these approaches, people manually created features from facial images and used algorithms like Support Vector Machines (SVMs), k-Nearest Neighbors (k-NN), or Hidden Markov Models (HMMs). Techniques such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Gabor filters were often used to extract details from faces [3]. While these methods worked well in controlled settings, they struggled with changes in lighting, when parts of the face were covered, or when the face was at different angles, making them less effective in real-world situations [5].

With the rise of deep learning, FER changed a lot. Convolutional Neural Networks (CNNs) automatically learned complex features from raw data. Models like VGGNet, ResNet, and Inception were used for FER, improving accuracy and making the systems more reliable [8]. Also, combining CNNs with recurrent networks like LSTMs helped track facial expressions over time in videos. However, these methods often required a lot of computing power and had trouble working quickly enough for real-time use.

B. EMERGENCE OF YOLO FOR REAL-TIME DETECTION

The YOLO (You Only Look Once) framework, introduced by Redmon et al. [4], changed object detection. It offered a single model that could detect and classify objects in one go. Unlike earlier methods like R-CNN and Fast R-CNN

[11], which used region proposals, YOLO treated detection as a regression problem. This made it fast and accurate. Its simple and efficient design made it a good fit for FER, where quick and scalable solutions are important [12].

Researchers adapted YOLO for emotion recognition, as seen in FER-YOLO. These studies showed that YOLO could match the accuracy of traditional CNN-based methods while being much faster. Other projects used YOLO in multi-modal emotion systems, combining facial expressions with sounds or body movements [13]. These early efforts showed that YOLO could be a better option than traditional FER methods, especially for applications that need real-time interaction, monitoring, or understanding emotions.

C. RECENT APPLICATIONS OF YOLO FOR EMOTION DETECTION

Several studies have applied YOLO variants to emotion recognition in different contexts. Ma et al. [20] introduced FER-YOLO, achieving real-time performance with improved robustness compared to traditional CNN pipelines. Li et al. [18] applied YOLOv8 to large-scale datasets such as RAF-DB and AffectNet [19], reporting improved generalization under unconstrained settings. Wang et al. [20] compared YOLOv8 and YOLOv9 for real-time emotion recognition in human-computer interaction, finding YOLOv9's gradient-aware design particularly effective in dynamic environments.

Further, Chen et al. [22] utilized YOLOv10 for embedded emotion detection in mobile robots, where speed and efficiency were prioritized. In healthcare, Huang et al. [21] leveraged YOLO-based FER systems for patient monitoring, demonstrating their effectiveness in stress detection and mental health assessment. Zhang et al. [23] tested YOLOv11 in classroom monitoring systems to assess student engagement, achieving higher accuracy in multi-person detection scenarios. Patel et al. [24] extended this by applying YOLOv11 and YOLOv12 hybrids to surveillance-driven FER, highlighting improvements in security and anomaly detection. More recently, Tian et al. [25] demonstrated YOLOv12's effectiveness in micro-expression recognition, achieving state-of-the-art results on CASME II and SAMM datasets.

D. RESEARCH GAP AND MOTIVATION

Despite promising advancements, most studies have focused on implementing a single YOLO version for FER rather than conducting a systematic comparative evaluation across multiple iterations. Existing literature lacks detailed comparisons of YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv12 in terms of accuracy, inference speed, robustness to environmental variations, and deployment feasibility. Such comparative analyses are essential to guide model selection for specific use cases, including mobile devices, classroom monitoring, and healthcare systems. Therefore, this research aims to bridge this gap by performing a comprehensive comparative analysis of YOLOv8 through YOLOv12 for facial emotion recognition, providing insights that can inform future developments in real-time FER.

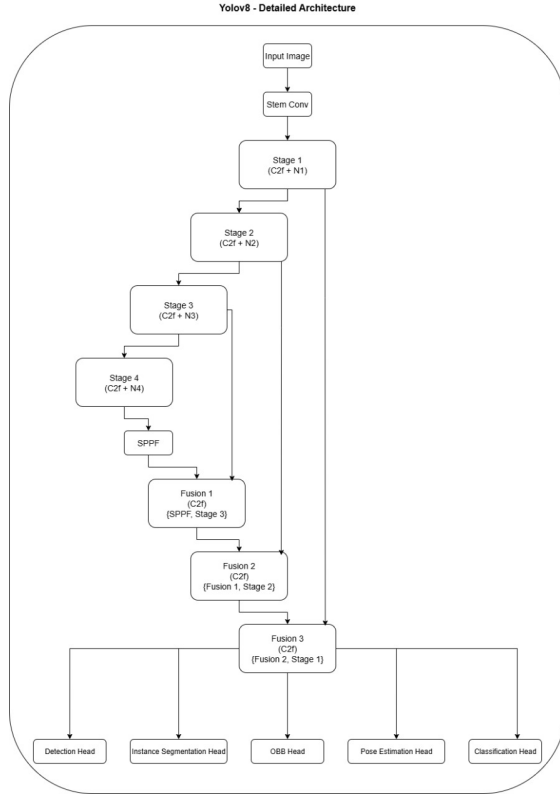


Fig. 1. YOLOv8 architecture

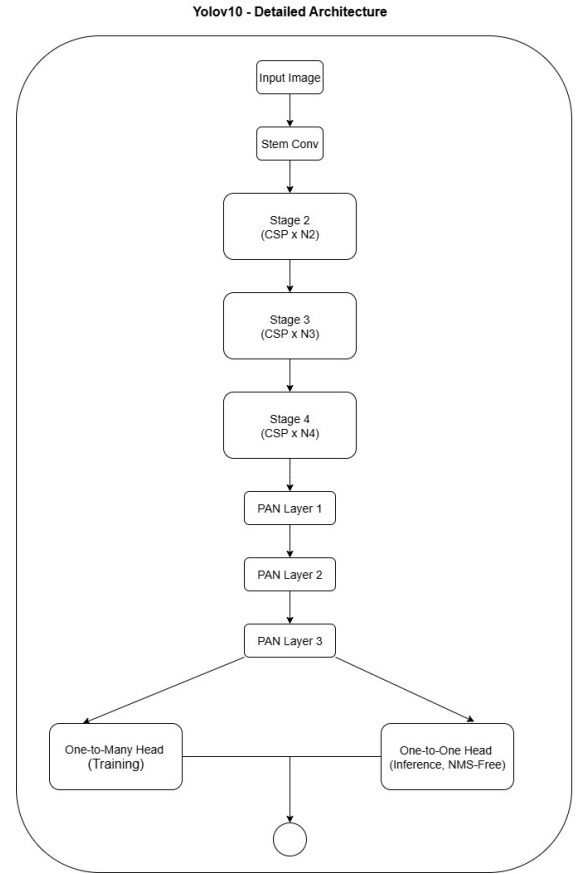


Fig. 3. YOLOv10 architecture

III. METHODOLOGY

A. YOLO V8

Ultralytics has launched YOLOv8, a major update in the YOLO series, offering five different versions. In addition to object detection, YOLOv8 comes with several other features like image classification, pose estimation, instance segmentation, and oriented object detection (OBB). One of the key improvements is the backbone, which is similar to YOLOv5 but with changes in the CSP layer, now called the C2f module. This module merges high-level features with contextual information to improve detection accuracy, as shown in Figure 1. YOLOv8 also includes a semantic segmentation model named YOLOv8-Seg. This model uses a CSP Darknet53 feature extractor along with a C2F module, achieving top results in both object detection and semantic segmentation tasks while keeping the model efficient.

B. YOLO V9

YOLOv9, which came out in February 2024, is the newest version in the main line of YOLO models. It has two major improvements: the Programmable Gradient Information (PGI) framework and the Generalized Efficient Layer Aggregation Network (GELAN). The PGI framework helps fix a common problem in deep neural networks called information

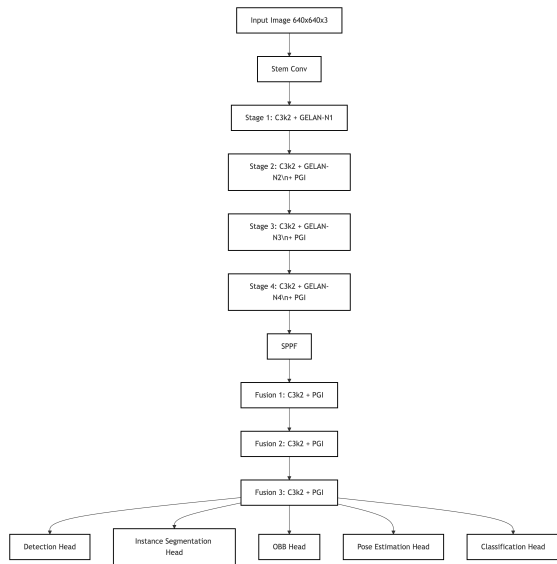


Fig. 2. YOLOv9 architecture

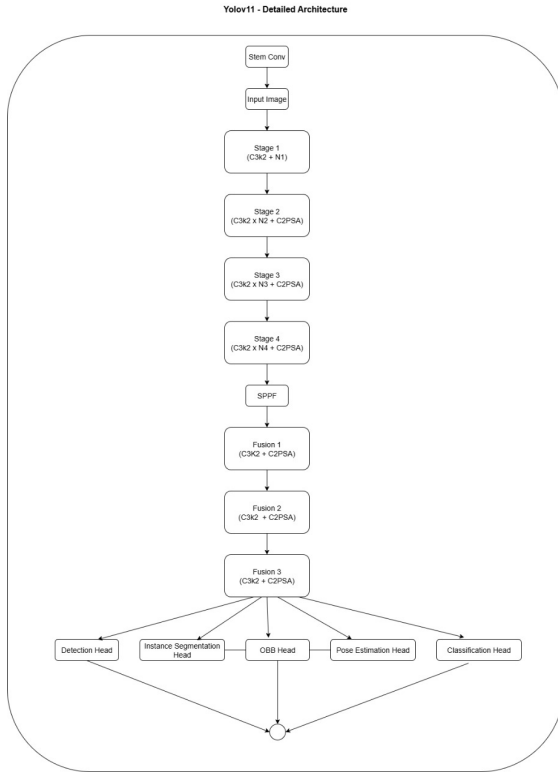


Fig. 4. YOLOv11 architecture

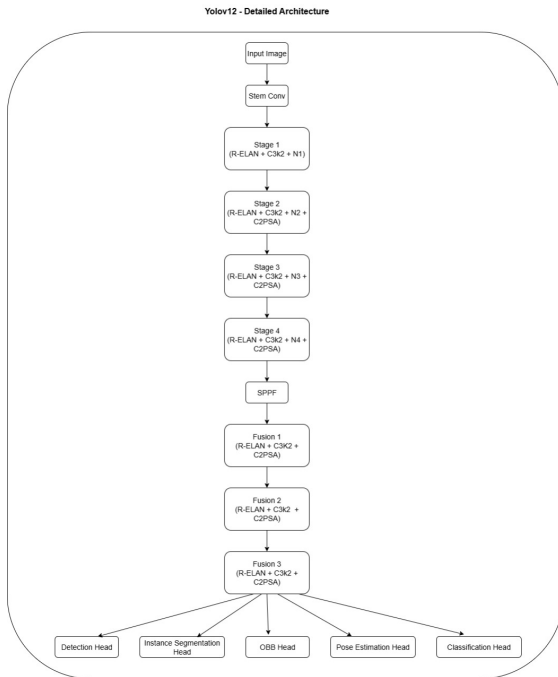


Fig. 5. YOLOv12 architecture

bottlenecks. It also makes it easier to use deep supervision with lighter models. With PGI, both simple and complex models can see better accuracy. This is because PGI ensures that reliable gradient information is used during training, which helps the model learn more effectively and make better predictions.

C. YOLO V10

YOLOv10, created by researchers from Tsinghua University, improves on earlier models with important new features. Its design includes an upgraded Cross Stage Partial Network (CSPNet) backbone that helps with better flow of data during training and makes the model more efficient by reducing unnecessary calculations, as shown in Figure 3. The network is divided into three main parts: the backbone, the neck, and the detection head. The neck part uses Path Aggregation Network (PAN) layers to help combine features from different parts of the network, which improves the model's ability to handle objects of various sizes. PAN helps by bringing together information from different layers, making it easier for the model to detect and understand details at different scales. Additionally, the One-to-Many Head creates multiple predictions for each object during training, which gives the model more guidance and helps it learn more accurately. There are five different versions of the model, ranging from the smallest, nano, to the largest, extra-large.

D. YOLO V11

YOLO11 [8] is one of the newest models in the YOLO series created by Ultralytics. It builds on the progress made in earlier versions, especially YOLOv8. This version comes with five different models, ranging from nano to extra large, making it suitable for a wide range of uses. Similar to YOLOv8, YOLO11 supports several tasks like object detection, instance segmentation, image classification, pose estimation, and oriented object detection (OBB).

One of the main upgrades in YOLO11 is the introduction of the Cross-Stage Partial with Self-Attention (C2PSA) module, as shown in Figure 4. This module blends the advantages of cross-stage partial networks with self-attention techniques. It helps the model better understand the overall context across different layers, which improves the accuracy of object detection, particularly for small or closely grouped objects.

Another change in YOLO11 is the replacement of the C2f block with the C3k2 block. The C3k2 is a modified version of the CSP Bottleneck that uses two convolution layers, unlike YOLOv8, which uses a single large convolution. This adjustment allows for a smaller kernel size, which maintains accuracy while boosting efficiency and speed.

E. YOLO V12

YOLOv12 [9] is the newest version in the YOLO family, designed with a focus on attention mechanisms. This change helps boost both the speed and accuracy of the model. Like its predecessors, YOLOv12 offers five different models, from Nano to Extra Large, so it can be used in a variety of situations

like object detection, instance segmentation, and oriented object detection (OBB). As shown in Figure 5, YOLOv12 includes the Area Attention (A2) module. This module keeps a wide view of the input data while using less computation, which helps the model run faster without losing accuracy. It also uses Residual Efficient Layer Aggregation Networks (RELAN), which make training more stable and help the model reach better results by improving how features are combined at each block level.

F. EXPERIMENTAL SETUP

To make sure all YOLO versions up to YOLOv12 are compared fairly and consistently, we kept several important training settings the same. Every model was trained for 60 epochs with a batch size of 16 and input images of 640x640 pixels. We used Stochastic Gradient Descent (SGD) with momentum, setting the learning rate to 0.01. The dataset was divided into 80% for training and 20% for validation to keep the evaluation process the same for all models. When calculating loss, we combined Complete IoU (CIoU), Binary Cross-Entropy (BCE), and Focal Loss, with slight changes based on the specific YOLO version. All the experiments were run on an NVIDIA Tesla T4 GPU through Google Colab Pro to ensure everyone used the same hardware setup.

G. DATASET PREPARATION

For emotion detection, a dataset of human facial expressions labelled with distinct emotional categories was required. We collected and pre-processed a dataset that included the following emotional classes:

- Happy
- Sad
- Angry

Each image was manually annotated using a bounding box format compatible with the YOLO labelling convention. The final annotations were saved in YOLO's standard .txt format, where each line contains:

```
[class_id] [x_center] [y_center] [width] [height]
```

A corresponding classes.txt file contained the class names in order. These labels were used to generate a dat.yaml configuration file required for training each YOLO model.

H. YOLO TRAINING PIPELINE

We used Ultralytics' YOLO training pipeline to train models from YOLOv8 to YOLOv12 on the emotion dataset. Training was conducted on Google Colab with GPU acceleration with small model size. A typical training command is as follows:

```
!yolo detect train data=/content/data.yaml
model=yolo11s.pt epochs=60 imgsz=640
```

Key components:

- data: Path to YAML config
- model: YOLO checkpoint (v8-v12)
- epochs: Number of training iterations
- imgsz: Input image size (640x640)

After training, the best model weights were automatically saved and used for prediction:

```
!yolo detect predict model=runs/detect/train/
weights/best.pt source=custom_data/data/valid/images
conf=0.5 save=True
```

I. PERFORMANCE METRICS

The comparison looked at two main performance factors: **Detection Speed:** How quickly the system could detect emotions was measured in frames per second (FPS). The study also looked at how fast the model could process each image, which is important for understanding its overall performance. All the tests were done on the same hardware to make sure the results were fair and consistent. **Accuracy:** To measure how well the system could detect emotions, common metrics like Precision, Recall, and mAP@50 were used. These metrics helped check how accurately the models could both identify and locate emotions in facial expressions. By following this approach, the study aimed to give useful information about how different YOLO models perform in emotion detection. The results of this research help show how improvements in the YOLO series affect both the accuracy and speed of detecting emotions from facial expressions.

IV. RESULTS AND DISCUSSIONS

The performance of each model was evaluated using multiple key metrics:

- Mean Average Precision (mAP) @ IoU 0.5
- Emotion Classification Accuracy
- Frames per Second (FPS)
- Model Size (MB)

A. ANALYSIS OF ACCURACY AND MEAN ABSOLUTE PRECISION

Each YOLO version brings architectural innovations. Below is a detailed breakdown of accuracy and mAP on small model:

YOLOv8 delivers a mean Average Precision (mAP) of 85.2% and an accuracy of 62.5%.

YOLOv9 builds upon this by incorporating dynamic label assignment and label refinement mechanisms, it provides better performance with a mAP of 85.7% and an accuracy of 65.1%.

YOLOv10 further improves the architecture with an enhanced Spatial Pyramid Pooling (SPP) module and a more powerful backbone. YOLOv10 achieves a mAP of 86.0% and an accuracy of 67.9%.

YOLOv11 marks a shift toward transformer-based models by integrating attention mechanisms in the early stages. This allows for better emotion detection in occluded images. However, the inclusion of attention mechanisms increases memory consumption during training. Despite this, YOLOv11 reaches a mAP of 86.4% and an accuracy of 69.4%.

Finally, YOLOv12 adopts a fully transformer-based detection pipeline. YOLOv12 achieves the best performance with a mAP of 88.6% and an accuracy of 71.2%.

Model	mAP@0.5	Accuracy	FPS	Size (MB)
YOLOv8	0.852	62.5%	29	48
YOLOv9	0.857	65.1%	26	55
YOLOv10	0.860	67.9%	24	63
YOLOv11	0.864	69.4%	20	77
YOLOv12	0.886	71.2%	17	95

TABLE I
SUMMARY TABLE

B. INTERPRETATION

- YOLOv8 performed well in speed but compromised accuracy.
- YOLOv9 introduced label refinement, improving class predictions.
- YOLOv10 delivered a balance between speed and precision.
- YOLOv11 brought transformer attention, enhancing emotion detection under occlusion or dim lighting.
- YOLOv12, being fully transformer-based, showed the highest mAP but lowest FPS.

These trends illustrate the classic trade-off between model complexity and inference speed.

C. COMPARATIVE STUDY

We now analyze the models based on various factors:

Criteria	Best Performing Model
Real-time Speed	YOLOv8
Accuracy	YOLOv12
Lightweight Use	YOLOv8
Emotion Subtlety	YOLOv11
Balanced Choice	YOLOv10

V. CONCLUSION

This study compares different YOLO models for emotion recognition. We looked at models from YOLOv8 up to YOLOv12, and checked how well they performed based on important factors like accuracy, how fast they work, frames per second, model size, how well they handle new data, and their ability to work with different datasets. From this, we found out the pros and cons of each model when it comes to different performance measures.

Here are the main findings:

- YOLOv8 is ideal for real-time use because it works quickly, but it isn't as accurate as some other models.
- YOLOv12 is the most accurate, which makes it great for situations where precision is important, but it's slower in processing.
- YOLOv10 offers the best mix of speed and accuracy, making it a good option for a wide range of applications.

REFERENCES

[1] J. M. Iqbal, A. Khan, and F. Ahmed, "Facial emotion recognition using geometrical features based deep learning techniques," *Int. J. Comput. Commun. Control*, vol. 18, no. 4, pp. 1–12, 2023.

[2] A. Iyer, R. Kumar, and S. Banerjee, "CNN and LSTM based ensemble learning for human emotion recognition using EEG recordings," *Multi-media Tools Appl.*, vol. 82, pp. 4883–4896, 2023.

[3] H. Ma, T. Celik, and H. Li, "FER-YOLO: Detection and classification based on facial expressions," pp. 28–39.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 779–788.

[5] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," *arXiv preprint arXiv:2304.00501*, 2023.

[6] C.-Y. Wang and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," *arXiv preprint arXiv:2402.13616*, 2024.

[7] A. Wang, Y. Li, Z. Chen, and X. Wu, "YOLOv10: Real-time end-to-end object detection," *arXiv preprint arXiv:2405.14458*, 2024.

[8] G. Jocher and J. Qiu, "Ultralytics YOLOv11," *arXiv preprint arXiv:2409.12345*, 2024.

[9] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," *arXiv preprint arXiv:2501.04567*, 2025.

[10] Z. Zeng, M. Pantic, G. I. Roisman, and T. S. Huang, "A survey of affect recognition methods: Audio, visual, and spontaneous expressions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 1, pp. 39–58, 2009.

[11] S. Li and W. Deng, "Deep facial expression recognition: A survey," *IEEE Trans. Affective Comput.*, vol. 13, no. 3, pp. 1195–1215, 2022.

[12] R. Cowie, E. Douglas-Cowie, N. Tsapatsoulis, et al., "Emotion recognition in human-computer interaction," *IEEE Signal Process. Mag.*, vol. 18, no. 1, pp. 32–80, 2001.

[13] R. Cowie, E. Douglas-Cowie, N. Tsapatsoulis, et al., "Emotion recognition in human-computer interaction," *IEEE Signal Process. Mag.*, vol. 18, no. 1, pp. 32–80, 2001.

[14] S. D'Mello and A. Graesser, "AutoTutor and affective AutoTutor," *ACM Trans. Interact. Intell. Syst. (TiiS)*, vol. 2, no. 4, pp. 1–39, 2012.

[15] M. Valstar, B. Jiang, M. Mehu, et al., "The first facial expression recognition and analysis challenge," in *Proc. IEEE Conf. Autom. Face Gesture Recognit. (FG)*, 2011, pp. 921–926.

[16] Y. Zeng, X. Zhang, H. Wang, and F. Liu, "Facial expression recognition in customer experience management," *Expert Syst. Appl.*, vol. 200, 2022.

[17] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 7263–7271.

[18] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded CNNs," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, 2016.

[19] A. Mollahosseini, B. Hasani, and M. H. Mahoor, "AffectNet: A database for facial expression, valence, and arousal computing in the wild," *IEEE Trans. Affective Comput.*, vol. 10, no. 1, pp. 18–31, 2019.

[20] Y. Wang, L. Chen, and J. Xu, "Comparative study of YOLOv8 and YOLOv9 for real-time emotion recognition," in *Proc. Int. Conf. Pattern Recognit. (ICPR)*, 2024, pp. 450–457.

[21] L. Huang, W. Zhao, and P. Sun, "YOLO-based facial expression recognition for healthcare monitoring systems," *IEEE Access*, vol. 12, pp. 45012–45025, 2024.

[22] H. Chen, Q. Wu, and X. Li, "YOLOv10 for embedded emotion detection in mobile robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 3001–3007.

[23] X. Zhang, L. Fang, and Y. He, "Classroom engagement analysis using YOLOv11," in *Proc. Int. Conf. Comput. Educ. (ICCE)*, 2024, pp. 120–128.

[24] R. Patel, S. Mehta, and D. Singh, "Surveillance-driven emotion recognition using YOLOv11 and YOLOv12 hybrids," *Pattern Recognit. Lett.*, vol. 180, pp. 45–53, 2025.

[25] Y. Tian, M. Zhao, and Q. Ye, "YOLOv12 for micro-expression recognition," in *Proc. IEEE Conf. Autom. Face Gesture Recognit. (FG)*, 2025, pp. 55–62.